A Constructive Introduction to Quantum Field Theory

Lecture 3: Stochastic observables, the Polchinski flow and forward–backward equations

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Abstract

We decompose the Gaussian free field along a scale parameter and take as primary objects the *stochastic observables*: martingales in the filtration of finite-resolution measurements. Interactions are introduced through the Polchinski equation for the effective potential. Its probabilistic counterpart is a forward–backward stochastic differential equation (FBSDE) for the field seen at increasing resolution, whose effective force is a martingale. Writing the FBSDE in the coordinate given by the conditional mean of the terminal field removes its drift and integrates the transport part of the force flow exactly. In the gradient case the remaining equation for the force is the Wetterich equation for the effective average action, whose propagator is the conditional covariance. We then discuss the variational structure of the flow (Boué–Dupuis formula and Hamilton–Jacobi–Bellman equation) in finite and infinite volume. The additional material covers Wick powers as observables in every dimension and their products (operator product expansion), the algebraic and perturbative structure of the flow (semigroup, flatness, counterterms for Φ^4_3), and the Dyson–Schwinger equations at every scale. The treatment is formal: we compute as if all objects were smooth and integrable, and we indicate where the real difficulties lie.

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Introduction

In Lectures 1–2 a Euclidean field was a probability measure on $\mathcal{S}'(\mathcal{M})$, i.e. a random distribution φ, and local operators were functions of φ. This viewpoint hides the scale-by-scale organisation of the Wilson–Kadanoff picture [WK74], and forces us to work with singular objects from the start. Already for the GFF, the Wick power $:\varphi^n:$ is not a random distribution when $n(d-2) \geq d$, since its regularisations are not even tight (Proposition 3.11 of Lecture 2).

In this lecture we index everything by a *resolution* $t \in [0, \infty)$. The field observed at resolution t is a smooth function φ_t, and the information available at resolution t is the σ-algebra $\mathcal{F}_t$ generated by $(\varphi_s)_{s \leq t}$. A physical quantity is then not a single random variable but a *consistent family* of finite-resolution estimates of it, i.e. a martingale. This is the notion of stochastic observable of [BCG26], on which this lecture is based; see also [Pol84, BK87, Sal99, BBD24] for the renormalisation group side and [BG20, BG23] for the variational side.

Structure. Part I (§§4–7) is the core of the lecture: the setting, the Polchinski equation and its FBSDE, the FBSDE in force coordinates and the Wetterich equation, and the variational structure. Part II (§§8–10) is additional material, referred to from Part I where needed.

What is formal. Rigorous statements require a finite-dimensional setting: a finite lattice, or a Fourier cutoff on the torus, together with a finite terminal resolution T. There, all the identities below are theorems under mild growth assumptions on the interaction, and their proofs are the computations we give. The hard problems are the limits $T \to \infty$ (ultraviolet)

and infinite volume (infrared), together with the locality of the limiting objects. We mark with (formal) the statements that we only derive at this level.

Conventions. As in Lectures 1–2: $\mathcal{M} = \mathbb{T}_L^d$ or $\mathbb{R}^d$, $A = m^2 - \Delta$, $C = A^{-1}$ with kernel G, μ_m the GFF, $\langle f, g\rangle = \int fg$ for real functions. For a functional F on smooth fields, $DF(\varphi) \in \mathcal{S}'(\mathcal{M})$ and $D^2F(\varphi)$ are its first and second derivatives, $D_hF = \langle DF, h\rangle$. $H_n(a;c)$ are the Hermite polynomials of variance c, and μ^Q is the centred Gaussian measure with covariance Q.

Part I. Main lecture

4 Setting: covariance profiles and observables

4.1 Covariance profiles

Definition 4.1. A *covariance profile* is a family $(Q_t)_{t\in[0,\infty)}$ of bounded, symmetric, translation invariant operators on $L^2(\mathcal{M})$ with $Q_0 = 0$, $t \mapsto Q_t$ smooth, $\dot{Q}_t := \partial_t Q_t \geq 0$, $Q_t \to C$ as $t \to \infty$, and such that $\dot{Q}_t$ and Q_t have smooth kernels for every $t < \infty$.

Our standard example is the *heat profile*

$$Q_t := C\,\mathrm{e}^{-A/t^2}, \qquad \dot{Q}_t = \frac{2}{t^3}\,\mathrm{e}^{-A/t^2}, \tag{4.1}$$

which smooths the Green function with the heat kernel at time t^{-2}. On $\mathbb{R}^d$ the kernels are explicit,

$$\dot{Q}_t(x-y) = \frac{2t^{d-3}\mathrm{e}^{-m^2/t^2}}{(4\pi)^{d/2}}\,\mathrm{e}^{-t^2|x-y|^2/4}, \tag{4.2}$$

so $\dot{Q}_t$ is concentrated on distances $\lesssim 1/t$, and in Fourier space Q_t retains the modes $|k| \lesssim t$: t is a *momentum scale*, or resolution. Other profiles are useful, even when they violate some smoothness requirement of Definition 4.1. A sharp Fourier cutoff is $\hat{Q}_t(k) = \mathbb{1}_{|k|\leq t}(|k|^2 + m^2)^{-1}$, which is not smooth in t. In a *finite range decomposition*, $\dot{Q}_t(x,y) = 0$ for $|x-y| \geq 1/t$ (Remark 8.4). We write

$$c_t := Q_t(x,x), \qquad Q_{T,t} := Q_T - Q_t \quad (t \leq T).$$

For $d \geq 2$, $c_t \to \infty$ as $t \to \infty$ (Exercise 4.2).

Let $(W_t)_{t\geq 0}$ be a cylindrical Brownian motion on $L^2(\mathcal{M})$, $\mathbb{E}[W_t(x)W_s(y)] = (t\wedge s)\delta(x-y)$, and

$$X_t := \int_0^t \dot{Q}_s^{1/2}\,\mathrm{d}W_s. \tag{4.3}$$

Proposition 4.2. *For every $t < \infty$, X_t is a smooth Gaussian field with law μ^{Q_t}, and $(X_t(x))_{t\geq 0}$ is a continuous martingale with $\mathrm{d}\langle X(x), X(y)\rangle_t = \dot{Q}_t(x,y)\,\mathrm{d}t$. As $t \to \infty$, $X_t \to X_\infty$ in H^{-s}_{loc} for $s > \frac{d}{2} - 1$, and $\mathrm{Law}(X_\infty) = \mu_m$. The increments $X_T - X_t$ are independent of $\mathcal{F}_t := \sigma(X_s : s \leq t)$, with law $\mu^{Q_{T,t}}$.*

So the GFF is the terminal value of a Gaussian martingale of smooth fields: the *field at resolution t* (Fig. 1). The proof is a variant of Theorem 3.2 of Lecture 2 and is left as Exercise 4.1.

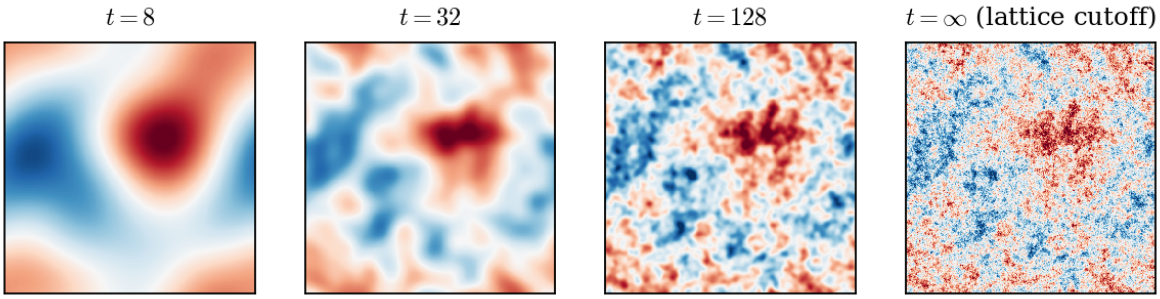


Figure 1: The field at resolution t, X_t, for the heat profile on $\mathbb{T}^2$ ($m = 4$, mean subtracted for display), at $t = 8, 32, 128$ and at the lattice cutoff, all built from the same white noise. The martingale X_t reveals finer and finer details, while the large scales are fixed at low resolution.

4.2 Stochastic observables

Fix a filtration $(\mathcal{F}_t)_{t\geq 0}$, for the moment that of X, and write $\mathbb{E}_t := \mathbb{E}[\,\cdot\,|\mathcal{F}_t]$.

Definition 4.3 ([BCG26])**.** An *observable* is a real or complex $(\mathcal{F}_t)$-martingale $O = (O_t)_{t\geq 0}$: $O_s = \mathbb{E}_s[O_t]$ for $s \leq t$. Its *expectation* is $\langle O\rangle := \mathbb{E}[O_t]$, which is independent of t. An *observable field* is a family $(O(x))_{x\in\mathcal{M}}$ of observables.

Every integrable $\mathcal{F}_\infty$-measurable random variable Y gives a *closed* observable $O_t = \mathbb{E}_t[Y]$, with $O_t \to Y$. The point of the definition is that interesting observables need *not* be closed: a martingale may diverge as $t \to \infty$ (for instance $X_t(x)$ for $d \geq 2$), and still carry a perfectly meaningful family of finite-resolution estimates. The next lemma is elementary but basic.

Lemma 4.4 (Asymptotic uniqueness)**.** *(a) If O^1, O^2 are observables with $\|O^1_t - O^2_t\|_{L^1} \to 0$, then $O^1 = O^2$. (b) If (N_t) is any adapted process and O an observable with $\|N_t - O_t\|_{L^1} \to 0$, then $O_s = \lim_{t\to\infty} \mathbb{E}_s[N_t]$ in L^1.*

Proof. (a) $O^1_s - O^2_s = \mathbb{E}_s[O^1_t - O^2_t]$ for every $t \geq s$, and conditional expectation is an L^1 contraction. (b) Likewise, $\|O_s - \mathbb{E}_s N_t\|_{L^1} \leq \|O_t - N_t\|_{L^1}$. □

So an observable is determined by its asymptotic behaviour. A *germ* for O is a family of functionals $\mathring{O}_t$ on smooth fields such that $\|O_t - \mathring{O}_t(X_t)\|_{L^1} \to 0$. The observable field $O(x)$ is *local* if it admits germs of the form

$$\mathring{O}_t(X_t)(x) = \mathring{\mathcal{O}}_t\big(X_t(x), \nabla X_t(x), \dots, \nabla^k X_t(x)\big).$$

Germs are not unique, and this freedom reflects the freedom of choosing counterterms. By Lemma 4.4(b), a germ determines the observable.

Exercises

Exercise 4.1 (The field at resolution t)**.** (a) Prove Proposition 4.2. *Hint:* on the torus, write $X_t = \sum_n (\int_0^t \dot{q}_n(s)^{1/2} \mathrm{d}B^n_s) e_n$. (b) For the heat profile, check $\int_0^\infty \dot{Q}_s \,\mathrm{d}s = C$ and $Q_t = C\mathrm{e}^{-A/t^2}$, and derive (4.2). (c) Show that $x \mapsto X_t(x)$ is a.s. real analytic for the heat profile.

Exercise 4.2 (Divergence of c_t)**.** For the heat profile on $\mathbb{R}^d$, show that $c_t = \frac{1}{4\pi} E_1(m^2/t^2) = \frac{1}{2\pi}\log\frac{t}{m} + O(1)$ for $d = 2$, where E_1 is the exponential integral, and that $c_t \asymp t^{d-2}$ for $d \geq 3$.

5 Interactions: the Polchinski equation and FBSDEs

5.1 Effective potentials

Fix a finite volume $\mathcal{M} = \mathbb{T}_L^d$ and a terminal resolution $T < \infty$, which plays the role of an ultraviolet cutoff: the reference field $X_T \sim \mu^{Q_T}$ is smooth. Let V_T be an interaction, a functional of smooth fields bounded below. Our main example is

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} \Big[\lambda H_4(\varphi(x); c_T) + \gamma_T \, H_2(\varphi(x); c_T) \Big] \mathrm{d}x, \tag{5.1}$$

with a mass counterterm γ_T to be fixed in Section 9.2 ($\gamma_T = 0$ for $d = 2$). The interacting measure at cutoff T is

$$\nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi). \tag{5.2}$$

Definition 5.1. The *effective potential* at resolution $t \in [0, T]$ is

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)} \, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{5.3}$$

V_t is the interaction felt by the field at resolution t once the fluctuations between t and T have been integrated out: it is the Wilson effective action. At $t = 0$, $V_0(0) = -\log Z_T$ is the free energy. Here is the first key fact.

Proposition 5.2 (Marginals)**.** *Let $\mathbb{P}$ be defined by $\mathrm{d}\mathbb{P} / \mathrm{d}\mathbb{P}_0 = Z_T^{-1} \mathrm{e}^{-V_T(X_T)}$, where $\mathbb{P}_0$ is the law of the Gaussian process X. Then $\mathrm{Law}_{\mathbb{P}}(X_T) = \nu_T$, and for every $t \le T$*

$$\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t, \qquad \nu_t(\mathrm{d}\varphi) := Z_T^{-1} \mathrm{e}^{-V_t(\varphi)} \mu^{Q_t}(\mathrm{d}\varphi). \tag{5.4}$$

Moreover, the conditional law of X_T given $\mathcal{F}_t$ is

$$\mu_t^{X_t}(\mathrm{d}\varphi), \qquad \mu_t^{\phi}(G) := \mathrm{e}^{V_t(\phi)} \int G(\phi + \zeta) \, \mathrm{e}^{-V_T(\phi+\zeta)} \, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{5.5}$$

Proof. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with an independent increment of law $\mu^{Q_{T,t}}$. Hence $\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1} \mathbb{E}_0[G(X_t) \mathrm{e}^{-V_T(X_T)}] = Z_T^{-1} \mathbb{E}_0[G(X_t) \mathrm{e}^{-V_t(X_t)}]$. The conditional law follows in the same way. □

So at every resolution the field is again a Gibbs measure, with reference covariance Q_t and potential V_t. The measure-valued martingale $(\mu_t^{X_t})_t$ goes from ν_T at $t = 0$ to δ_{X_T} at $t = T$: it is a *stochastic localisation* of ν_T in the sense of Eldan [Eld23], adapted to the covariance profile [BBD24, BCG26].

Theorem 5.3 (Polchinski equation [Pol84])**.** (formal) *V_t solves the backward Hamilton–Jacobi equation*

$$\partial_t V_t = \tfrac{1}{2} \langle DV_t, \dot{Q}_t \, DV_t \rangle - \tfrac{1}{2} \operatorname{Tr}\big(\dot{Q}_t \, D^2 V_t \big), \qquad t \in [0, T], \tag{5.6}$$

with terminal condition V_T.

Proof. $u_t := \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2} \Delta_{Q_{T,t}}} u_T$ is a Gaussian convolution, where $\Delta_Q := \operatorname{Tr}(Q D^2)$. Since $\partial_t Q_{T,t} = -\dot{Q}_t$, it solves the backward heat equation $\partial_t u_t = -\frac{1}{2} \operatorname{Tr}(\dot{Q}_t D^2 u_t)$. Insert $D^2 u = (DV \otimes DV - D^2 V) u$ and divide by $-u$. □

The nonlinear term generates new, generally nonlocal, interactions (trees), and the trace term contracts pairs of fields with the covariance of the shell $[t, t + \mathrm{d}t]$ (loops). The substitution $u = \mathrm{e}^{-V}$, which linearises the equation, is a Cole–Hopf transform: we come back to this in Section 5.4, and in Section 9.

5.2 The Polchinski process

Theorem 5.4 (FBSDE for the interacting field)**.** (formal) *Under* $\mathbb{P}$ *the process* $\Phi_t := X_t$*, the* interacting field at resolution t*, solves*

$$\mathrm{d}\Phi_t = \dot{Q}_t\, F_t\, \mathrm{d}t + \dot{Q}_t^{1/2}\, \mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -DV_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\big[-DV_T(\Phi_T)\big], \tag{5.7}$$

with $\Phi_0 = 0$ *and a* $\mathbb{P}$*-cylindrical Brownian motion* $W^{\mathbb{P}}$*. In particular the* effective force $(F_t)_{t\le T}$ *is a* $\mathbb{P}$*-martingale: an observable field of the interacting theory.*

Proof. The density process is $M_t := \mathbb{E}_0[\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0|\mathcal{F}_t] = Z_T^{-1}\mathrm{e}^{-V_t(X_t)}$, by Proposition 5.2. By Itô's formula and (5.6), its drift vanishes and $\mathrm{d}M_t = -M_t\langle DV_t(X_t), \dot{Q}_t^{1/2}\mathrm{d}W_t\rangle$. By Girsanov's theorem, $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2} DV_s(X_s)\mathrm{d}s$ is a $\mathbb{P}$-Brownian motion, which is (5.7). Differentiating (5.3) gives $DV_t(\phi) = \mu_t^{\phi}(DV_T)$, which is the second identity. Alternatively, Itô's formula for $DV_t(\Phi_t)$, together with the derivative of (5.6), shows that its drift vanishes. Here one uses that D^2V_t is symmetric. □

Read as an equation for an unknown pair (Φ, F), (5.7) is a *forward–backward* system [MY99]. The field is driven forward in resolution by a force that is determined backward, as the conditional expectation of a functional of the terminal field:

$$\Phi_t = \int_0^t \dot{Q}_s F_s\, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2}\mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\big[-DV_T(\Phi_T)\big], \qquad t \in [0,T]. \tag{5.8}$$

Here and below we drop the superscript and write W for the Brownian motion driving the interacting field. The Polchinski equation is not needed to state (5.8), and $\nu_T = \mathrm{Law}(\Phi_T)$ is recovered from the solution. This is the starting point of FBSDE constructions of Euclidean fields (Fig. 2): one solves (5.8) with renormalised V_T, with estimates uniform in T and in the volume, and takes the limit. The field coordinate Φ is *not* an observable, only a semimartingale. Its martingale part is the Gaussian noise, and the observables are built from it and from F.

Remark 5.5 (Closed versus local). At finite T the force is a closed observable, $F_t = \mathbb{E}_t[F_T]$. For Φ^4, $F_T(x) = -4\lambda H_3(\Phi_T(x); c_T) - 2\gamma_T H_1(\Phi_T(x); c_T)$ is local. As $T \to \infty$ one expects the force to converge as an observable, but not to an integrable terminal value. Suppose the DS equation of Section 10 passed to the limit with some $F_\infty \in L^1$. Conditioning it on $\sigma(\Phi_\infty)$ would produce a function of Φ_∞ in the role of the renormalised cube, and the renormalised cube is not a random variable under the Φ_3^4 measure [ALZ06]. The limiting force is an observable in the proper sense: a compatible family with no terminal representative. Locality of the limiting force is the crux of the construction [BCG26].

5.3 Observables of the interacting theory

For a functional G of the terminal field, the observable $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)]$ is $O_t = G_t(\Phi_t)$, with

$$G_t(\phi) = \mu_t^{\phi}(G) = \mathrm{e}^{V_t(\phi)} \int G(\phi+\zeta)\, \mathrm{e}^{-V_T(\phi+\zeta)} \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{5.9}$$

Formally, G_t solves the backward Kolmogorov equation of the interacting dynamics,

$$\partial_t G_t + \tfrac{1}{2}\,\mathrm{Tr}\big(\dot{Q}_t D^2 G_t\big) - \langle \dot{Q}_t DV_t, DG_t\rangle = 0, \qquad G_T = G, \tag{5.10}$$

which is the Doob transform of the Gaussian heat equation by e^{-V_t} (Exercise 5.3). Conversely, any solution of (5.10) yields an observable $G_t(\Phi_t)$; at finite T it is closed, but as $T \to \infty$ it need not be. The force is the case $G = -DV_T$, and is special because it also drives the dynamics.

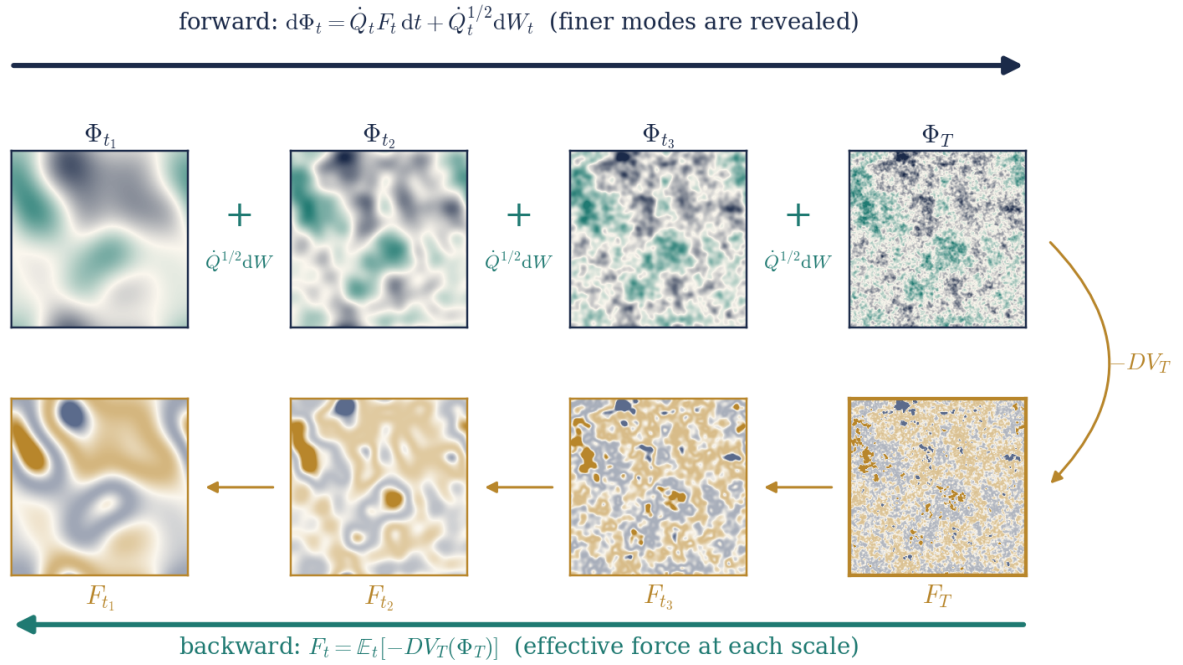


Figure 2: The forward–backward structure of (5.8) on $\mathbb{T}^2$, heat profile, at resolutions $t_1 < t_2 < t_3 < T$, all built from the same noise. Top: the field gains the fluctuation of each shell, $\dot{Q}_t^{1/2}\mathrm{d}W_t$, and reveals finer modes. Bottom: the effective force $F_t = \mathbb{E}_t[-DV_T(\Phi_T)]$ for the Φ^4 interaction, determined backward from the terminal interaction. It is shown to first order in λ, where it is the Wick cube $-4\lambda H_3(\Phi_t; c_t)$, and each panel is in its own units.

Tangent observables. Let V_T^θ depend on a coupling θ, and let $U_t := \partial_\theta V_t^\theta$. Differentiating (5.3) gives $U_t(\Phi_t) = \mathbb{E}_t^\theta[\partial_\theta V_T^\theta(\Phi_T)]$, an observable, and

$$\partial_\theta \mathbb{E}^{\nu_T^\theta}[G] = -\operatorname{Cov}^{\nu_T^\theta}\big(G, \partial_\theta V_T^\theta\big). \tag{5.11}$$

Thus the tangent observables, such as the running $:\varphi^4:$ for $\theta = \lambda$ (together with the derivative of the counterterm $\partial_\lambda \gamma_T \int :\varphi^2:$), describe the response of the theory to a change of coupling. They are the observables conjugate to the coordinates of theory space.

5.4 The force and a Burgers equation

Differentiating (5.6), the effective force $F_t = -DV_t$, now as a function of the field, solves

$$\partial_t F_t = -DF_t[\dot{Q}_t F_t] - \tfrac{1}{2}\operatorname{Tr}\big(\dot{Q}_t D^2 F_t\big), \qquad F_T = -DV_T, \tag{5.12}$$

an infinite-dimensional, backward, viscous *Burgers equation*. The Cole–Hopf transform $F = D\log u$ linearises it, which is the statement that $u = \mathrm{e}^{-V}$ solves a heat equation. Two remarks on (5.12):

(i) it uses only the force, which is a local quantity (*intensive*), while V_t is *extensive*, of order of the volume. This matters in infinite volume (Section 7.2);

(ii) the Burgers flows for different covariances commute, and the curvature vanishes, when DF_t is symmetric, i.e. when F_t is a gradient [BCG26]. For a non-gradient force, as for the general Wilson–Itô diffusions of [BCG26], the flows do not commute.

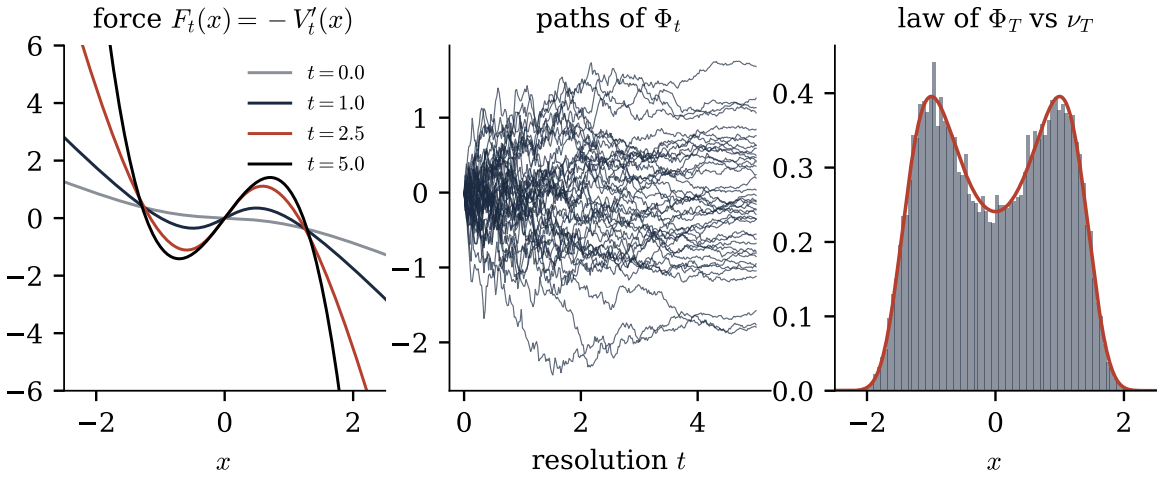


Figure 3: A zero-dimensional Polchinski process (Exercise 5.2): $Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: the effective force $F_t = -V_t'$ is almost linear at low resolution and develops the double-well structure as $t \to T$. Centre: paths of the FBSDE (5.7), which split between the two wells (stochastic localisation). Right: histogram of Φ_T ($2 \cdot 10^4$ paths) against the density of ν_T.

Exercises

Exercise 5.1 (Quadratic interactions)**.** Let $V_T(\varphi) = \frac{\kappa}{2}\langle\varphi, \varphi\rangle$, $\kappa > 0$. (a) Show that $V_t(\varphi) = \frac{1}{2}\langle\varphi, \kappa(1 + \kappa Q_{T,t})^{-1}\varphi\rangle + \text{const}$ solves (5.6). (b) Show that ν_T is Gaussian with covariance $(Q_T^{-1} + \kappa)^{-1}$: for $Q_T \to (m^2 - \Delta)^{-1}$ this is the GFF of mass $(m^2 + \kappa)^{1/2}$. (c) Write (5.7) explicitly. It is a linear FBSDE. Solve it, and check that F_t is a martingale. *Moral:* the mass can be put either into the covariance or into the force.

Exercise 5.2 (Zero dimensions)**.** Take $\mathcal{M}$ a point, $Q_t = q(t)$ scalar, increasing, $V_T(x) = \lambda x^4 - ax^2$ with $\lambda > 0$, $a \geq 0$ (a double well if $a > 0$; Fig. 3 has $\lambda = \frac{1}{2}$, $a = \frac{3}{2}$). (a) Write (5.6) and solve it numerically for $q(t) = 1 - \mathrm{e}^{-t}$. (b) Simulate (5.7) by first computing V_t' on a grid, and compare the histogram of Φ_T with $\mathrm{e}^{-\lambda x^4 + ax^2 - x^2/2q(T)}$. (c) Compute V_t to second order in λ.

Exercise 5.3 (Doob transform)**.** Show that G_t in (5.9) solves (5.10), by writing $\mathrm{e}^{-V_t}G_t = \mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}}(\mathrm{e}^{-V_T}G)$. Prove (5.11).

6 The FBSDE in force coordinates: the Wetterich equation

In this section the forward–backward system (5.8) is the primary object. We look for the coordinates in which it is simplest. They remove the predictable part of the forward dynamics, and they integrate exactly the bilinear (tree) part of the force flow (5.12). In the gradient case the resulting equation for the force is the Wetterich equation for the effective average action [Wet93, BTW02]. Throughout,

$$S_t := Q_T - Q_t, \qquad \dot{S}_t = -\dot{Q}_t, \qquad S_T = 0,$$

is the covariance still unresolved at resolution t, and (Φ, F) solves (5.8) with a martingale force $F_t = F_t(\Phi_t)$, $\mathrm{d}F_t = \Upsilon_t\,\mathrm{d}W_t$.

6.1 State and costate

The FBSDE is the stochastic Pontryagin system of the control problem of Theorem 7.1. The controlled field $\mathrm{d}X = \dot{Q}_t^{1/2}(u\,\mathrm{d}t + \mathrm{d}W)$ has Hamiltonian $\langle p, \dot{Q}_t^{1/2}u\rangle - \frac{1}{2}\|u\|^2$, which does not depend on the state. By the stochastic maximum principle (Exercise 6.1), the adjoint process p has no drift, $\mathrm{d}p_t = \mathrm{Y}_t\,\mathrm{d}W_t$, with terminal value $p_T = -DV_T(X_T)$, and the optimal control is $u_t = \dot{Q}_t^{1/2}p_t$. Thus

the field Φ is the state, and the force F is the costate.

The function $F_t(\cdot)$ with $F_t = F_t(\Phi_t)$ is the *decoupling field* of the FBSDE, and the Burgers equation (5.12) is its quasilinear PDE; $\mathrm{Y}_t = DF_t(\Phi_t)\dot{Q}_t^{1/2}$ is the martingale representation integrand of the costate. The two terms of (5.12) have distinct origins. The transport term $-DF[\dot{Q}F]$ comes from the drift of the state, and generates the tree diagrams. The diffusion term $-\frac{1}{2}\operatorname{Tr}(\dot{Q}D^2F)$ is the Itô correction, and generates the loops.

Without noise the state–costate pair follows the Hamiltonian flow of the free Hamiltonian $h_s(p) = \frac{1}{2}\langle p, \dot{Q}_s p\rangle$: $\dot{p} = 0$ and $\dot{x} = \dot{Q}_s p$. From t to T this flow is the linear shear

$$\Sigma_t : (x, p) \longmapsto (x + S_t p,\ p). \tag{6.1}$$

Its trajectories are the characteristics of the transport part of (5.12) (Exercise 6.2).

6.2 Removing the predictable drift

The drift $\dot{Q}_t F_t$ of the state is predictable, and the costate is a martingale. Hence the drift that remains to be accumulated on $[t, T]$ is predicted, at resolution t, by transporting the current state with the current costate along the free flow (6.1).

Proposition 6.1 (The forecast)**.** (formal) *The conditional mean of the terminal field is*

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \tag{6.2}$$

i.e. $(\psi_t, F_t) = \Sigma_t(\Phi_t, F_t)$*. It is a martingale, with*

$$\mathrm{d}\psi_t = \dot{Q}_t^{1/2}\mathrm{d}W_t + S_t\mathrm{Y}_t\,\mathrm{d}W_t = \big(\mathbb{1} + S_t DF_t(\Phi_t)\big)\dot{Q}_t^{1/2}\mathrm{d}W_t. \tag{6.3}$$

Proof. Integrate the forward equation from t to T and condition on $\mathcal{F}_t$: since F is a martingale, $\mathbb{E}_t\int_t^T \dot{Q}_s F_s \mathrm{d}s = S_t F_t$. For the differential, the drift $\dot{Q}_t F_t$ of Φ cancels against $\dot{S}_t F_t$, and $S_t\,\mathrm{d}F_t = S_t\mathrm{Y}_t\,\mathrm{d}W_t$. □

So ψ is a closed observable, the *field observable* of [BCG26]. The field splits as

$$\Phi_t = \psi_t - S_t F_t : \tag{6.4}$$

a martingale, minus the drift still expected on the unresolved scales, which vanishes at $t = T$. The noise of ψ in (6.3) has two sources: the fresh fluctuation of the shell, $\dot{Q}_t^{1/2}\mathrm{d}W_t$, and the revision of the forecast of the remaining drift, $S_t\,\mathrm{d}F_t$. The coupling between the forward and the backward equations, which was in the drift of Φ, now sits in the volatility of ψ. The coordinate ψ is the *interaction picture* with respect to the noiseless (classical) part of the dynamics.

6.3 The FBSDE in the forecast coordinate

Assume that the map $x \mapsto \psi = x + S_t F_t(x)$ is invertible, and write its inverse as

$$x = \psi + S_t \mathcal{F}_t(\psi), \qquad \mathcal{F}_t(\psi) := -F_t(x), \tag{6.5}$$

so that $F_t = -\mathcal{F}_t(\psi_t)$: the costate is now a function of the forecast. Introduce the *response* and the *regulator*

$$H_t(\psi) := \big(S_t^{-1} + D\mathcal{F}_t(\psi)\big)^{-1}, \qquad \mathcal{R}_t := S_t^{-1} - Q_T^{-1}, \qquad \dot{\mathcal{R}}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geq 0. \tag{6.6}$$

Differentiating the two maps $x \leftrightarrow \psi$ gives $\mathbb{1} + S_t DF_t(x) = (\mathbb{1} + S_t D\mathcal{F}_t(\psi))^{-1} = H_t(\psi) S_t^{-1}$.

Theorem 6.2 (FBSDE in force coordinates)**.** (formal) *In the coordinates $(\psi, \mathcal{F})$, the system (5.8) becomes*

$$\mathrm{d}\psi_t = H_t(\psi_t)\, S_t^{-1} \dot{Q}_t^{1/2}\, \mathrm{d}W_t, \qquad \mathcal{F}_t(\psi_t) = \mathbb{E}_t\big[DV_T(\psi_T)\big], \tag{6.7}$$

with $\psi_0 = S_0 F_0 = \mathbb{E}[\Phi_T]$ and $\psi_T = \Phi_T$. The forward equation has no drift, and its covariance is $\mathcal{C}_t := H_t \dot{\mathcal{R}}_t H_t^*$. The decoupling field $\mathcal{F}$ solves the pure diffusion equation

$$\partial_t \mathcal{F}_t + \tfrac{1}{2}\,\mathrm{Tr}\big(\mathcal{C}_t\, D^2 \mathcal{F}_t\big) = 0, \qquad \mathcal{F}_T = DV_T. \tag{6.8}$$

Proof. The forward equation is (6.3) with $\mathbb{1} + S_t DF_t = H_t S_t^{-1}$, and its covariance is

$$H_t S_t^{-1} \dot{Q}_t S_t^{-1} H_t^* = H_t \dot{\mathcal{R}}_t H_t^*.$$

The backward equation is $F_t = \mathbb{E}_t[F_T]$ with $\psi_T = \Phi_T$. Since $\mathcal{F}_t(\psi_t)$ is a martingale and ψ has no drift, the drift in Itô's formula for $\mathcal{F}_t(\psi_t)$ is the left-hand side of (6.8). □

Compare (6.8) with the Burgers equation (5.12): the transport term has been absorbed exactly by the change of coordinates, and only the Itô term is left. If it is dropped, $\mathcal{F}_t = DV_T$ does not move: all the tree diagrams are contained in the coordinate change (6.5). The loops are generated only by the quadratic variation $\mathcal{C}_t$ of the forecast, i.e. by the randomness of the noise and of the force revisions, and the nonlinearity has moved into the field dependence of $\mathcal{C}_t$. None of this uses a gradient structure or a Gaussian convolution: Theorem 6.2 holds for any FBSDE of the form (5.8) with a Markov martingale force, gradient or not, for which the coordinate change is invertible. For a general terminal force $F_T = \mathfrak{f}(\Phi_T)$, replace DV_T by $-\mathfrak{f}$ in (6.7) and (6.8).

6.4 Conditional moments

For the Polchinski process the response has a direct meaning.

Proposition 6.3. (formal) *If $F_t = -DV_t$ then*

$$H_t(\psi_t) = \mathrm{Cov}_t := \mathrm{Cov}^{\mathbb{P}}(\Phi_T \,|\, \mathcal{F}_t) = S_t - S_t D^2 V_t(\Phi_t) S_t.$$

In particular $H_t > 0$ is symmetric, $\mathbb{1} + S_t DF_t = \mathrm{Cov}_t\, S_t^{-1}$ is always invertible, and $H_t \leq S_t$ when V_T is convex.

Proof. By (5.5), the conditional law μ_t^{ϕ} is the tilt of $\mu^{S_t}(\cdot - \phi)$ by e^{-V_T}; differentiating twice in ϕ gives $\mathrm{Cov}(\mu_t^{\phi}) = S_t - S_t D^2 V_t(\phi) S_t$. On the other hand $H_t S_t^{-1} = \mathbb{1} + S_t DF_t = \mathbb{1} - S_t D^2 V_t$. The last claim is the Brascamp–Lieb inequality. □

So the forecast coordinates are the first two conditional moments of the stochastic localisation $\mu_t^{\Phi_t}$ of Proposition 5.2: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}_t\Phi_T, \mathrm{Cov}_t\,\Phi_T)$ (Fig. 4). They are tied by Itô's formula: since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle\psi,\psi\rangle_t = \mathcal{C}_t\,\mathrm{d}t = \mathrm{Cov}_t\,\dot{\mathcal{R}}_t\,\mathrm{Cov}_t\;\mathrm{d}t, \qquad \mathrm{d}\,\mathrm{Cov}_t = -\,\mathrm{Cov}_t\,\dot{\mathcal{R}}_t\,\mathrm{Cov}_t\;\mathrm{d}t + \mathrm{d}(\text{martingale}). \tag{6.9}$$

Conditional variance is resolved at the rate $\mathrm{Cov}_t\,\dot{\mathcal{R}}_t\,\mathrm{Cov}_t$, which for the Gaussian reference is $S_t\dot{\mathcal{R}}_tS_t = \dot{Q}_t$.

6.5 The gradient sector: Wetterich's equation

Now let $F_t = -DV_t$. The graph $\{(x, -DV_t(x))\}$ is a Lagrangian submanifold of the (field, force) phase space, and the shear (6.1) is symplectic because S_t is symmetric. Its image is again Lagrangian. It is a graph over ψ: the map $x \mapsto x + S_tF_t(x)$ is S_t times the gradient of $\frac{1}{2}\langle x, S_t^{-1}x\rangle - V_t(x)$, which is strictly convex since its Hessian is $S_t^{-1}\,\mathrm{Cov}_t\,S_t^{-1} > 0$ (Proposition 6.3), so the map is injective. Hence the image is the graph of a gradient:

$$\mathcal{F}_t = D\Gamma_t, \qquad H_t = \big(S_t^{-1} + D^2\Gamma_t\big)^{-1}, \qquad \Gamma_T = V_T. \tag{6.10}$$

The generating function of this canonical transformation is the Moreau (infimal convolution) transform

$$V_t(x) = \inf_{\psi}\Big\{\Gamma_t(\psi) + \tfrac{1}{2}\langle x-\psi, S_t^{-1}(x-\psi)\rangle\Big\}, \tag{6.11}$$

whose optimality conditions $x = \psi + S_tD\Gamma_t(\psi)$, $DV_t(x) = D\Gamma_t(\psi)$ are (6.5). It is well defined by the same convexity: up to the change of variable $j = S_t^{-1}x$, it is a Legendre transform. We use (6.11) to fix the additive constant in Γ_t. Figure 5 shows the construction in zero dimensions.

Theorem 6.4 (Wetterich equation)**.** (formal) Γ_t *solves*

$$\partial_t\Gamma_t = \tfrac{1}{2}\,\mathrm{Tr}\Big[\dot{\mathcal{R}}_t\Big(\big(S_t^{-1} + D^2\Gamma_t\big)^{-1} - S_t\Big)\Big], \qquad \Gamma_T = V_T. \tag{6.12}$$

Equivalently, the effective average action $\Gamma_t^{\mathrm{EAA}}(\psi) := \Gamma_t(\psi) + \frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle$ *solves, up to a field-independent term,*

$$\partial_t\Gamma_t^{\mathrm{EAA}} = \tfrac{1}{2}\,\mathrm{Tr}\Big[\dot{\mathcal{R}}_t\big(D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t\big)^{-1}\Big]. \tag{6.13}$$

It interpolates between the bare action $\Gamma_T^{\mathrm{EAA}} = \frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle + V_T(\psi)$ *($\mathcal{R}_T = \infty$) and the 1PI effective action* Γ_0^{EAA} *of* ν_T *($\mathcal{R}_0 = 0$).*

Proof. Integrate (6.8) in field space. With $\mathcal{F} = D\Gamma$, the derivative of $H = (S^{-1} + D^2\Gamma)^{-1}$ in a direction v is $-H\,D^3\Gamma[v]\,H$, so by the total symmetry of $D^3\Gamma$,

$$D_v\,\mathrm{Tr}\big(\dot{\mathcal{R}}_tH_t\big) = -\,\mathrm{Tr}\big(H_t\dot{\mathcal{R}}_tH_t\,D^3\Gamma_t[v]\big) = -\,\mathrm{Tr}\big(\mathcal{C}_t\,D^2\mathcal{F}_t\big)[v].$$

Hence (6.8) reads $D\big(\partial_t\Gamma_t - \frac{1}{2}\,\mathrm{Tr}(\dot{\mathcal{R}}_tH_t)\big) = 0$, and (6.12) holds up to a function of t alone. That this function is $-\frac{1}{2}\,\mathrm{Tr}(\dot{\mathcal{R}}_tS_t)$ with the normalisation (6.11) follows from the Polchinski equation by the envelope theorem (Exercise 6.5). For (6.13), note that $D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t = D^2\Gamma_t + S_t^{-1}$. For the endpoints: $S_T = 0$; and at $t = 0$, $S_0 = Q_T$, so (6.11) identifies $\frac{1}{2}\langle x, Q_T^{-1}x\rangle - V_0(x)$, up to a constant, with the cumulant generating function of ν_T at $Q_T^{-1}x$, whose Legendre transform is Γ_0^{EAA} (Exercise 6.7). □

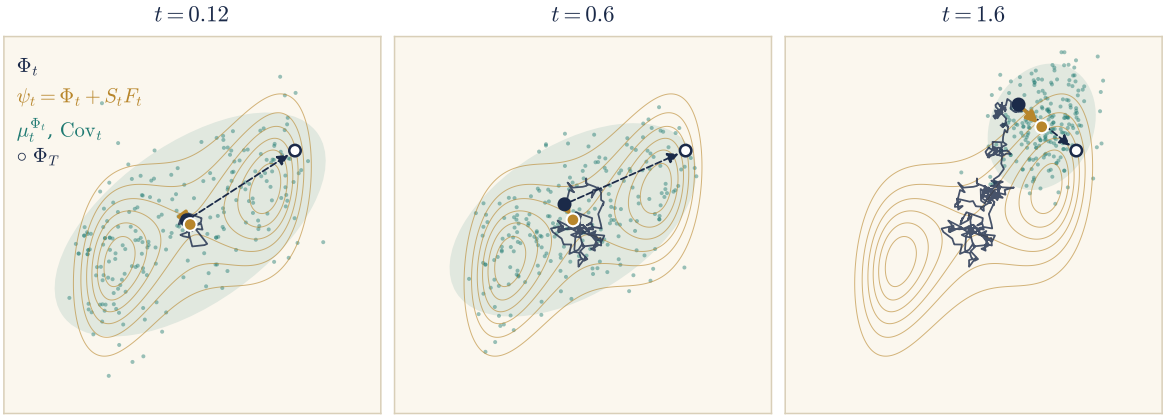


Figure 4: Stochastic localisation for a two-component field: $Q_t = (1 - \mathrm{e}^{-t})\mathbb{1}$, $T = 4$, V_T a tilted double well, and the contours show the density of ν_T. One path of (5.7) up to resolution t (navy), with its terminal value Φ_T (open circle). The forecast $\psi_t = \Phi_t + S_t F_t$ (gold) is the mean of the conditional law $\mu_t^{\Phi_t}$: teal samples, and the ellipse is two standard deviations of Cov_t. At low resolution the conditional law covers both wells; as t grows it localises on one of them.

Thus the Wetterich equation is the gradient-integrated form of the force equation (6.8), i.e. of the backward half of the FBSDE in forecast coordinates [Mor94, BCG26]. Every object in it is an observable of the Polchinski process:

Wetterich	FBSDE
field argument ψ	forecast $\psi_t = \mathbb{E}_t[\Phi_T]$, the state transported by Σ_t
$D\Gamma_t(\psi_t)$	minus the costate, $-F_t = \mathbb{E}_t[DV_T(\Phi_T)]$
propagator $(D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t)^{-1}$	conditional covariance Cov_t
regulator $\mathcal{R}_t = S_t^{-1} - Q_T^{-1}$	precision of the unresolved shell, minus that of the prior
$\partial_t \Gamma_t$	$\frac{1}{2}\,\mathrm{Tr}[\dot{\mathcal{R}}_t(\mathrm{Cov}_t - S_t)]$

The flow of Γ is driven by the difference between the interacting and the free conditional covariances, paired with the shell $\dot{\mathcal{R}}_t$. It is *one-loop exact*, with the full propagator inside the trace. In the Gaussian case $D^2\Gamma_t$ is constant and only the constant term of Γ flows. For the heat profile with $T = \infty$, $S_t = C(1 - \mathrm{e}^{-A/t^2})$ and

$$\mathcal{R}_t = \frac{A}{\mathrm{e}^{A/t^2} - 1}, \tag{6.14}$$

which behaves like a mass term t^2 on the modes $|k| \ll t$ (for $m \ll t$) and is exponentially small on the modes $|k| \gg t$: the exponential regulator of the functional renormalisation group (Fig. 6).

6.6 Variational meaning and the tree level

Constrained free energies. (formal) By the Gibbs variational principle applied to the conditional law (5.5), and the identity $\mathrm{Ent}(\nu|\mu^S(\cdot - x)) = \mathrm{Ent}(\nu|\mu^S(\cdot - \psi)) + \frac{1}{2}\langle x - \psi, S^{-1}(x - \psi)\rangle$ for ν of mean ψ,

$$\Gamma_t(\psi) = \inf\Big\{ \int V_T \,\mathrm{d}\nu + \mathrm{Ent}\big(\nu \,|\, \mu^{S_t}(\cdot - \psi)\big) \,:\, \int \varphi\, \nu(\mathrm{d}\varphi) = \psi \Big\}. \tag{6.15}$$

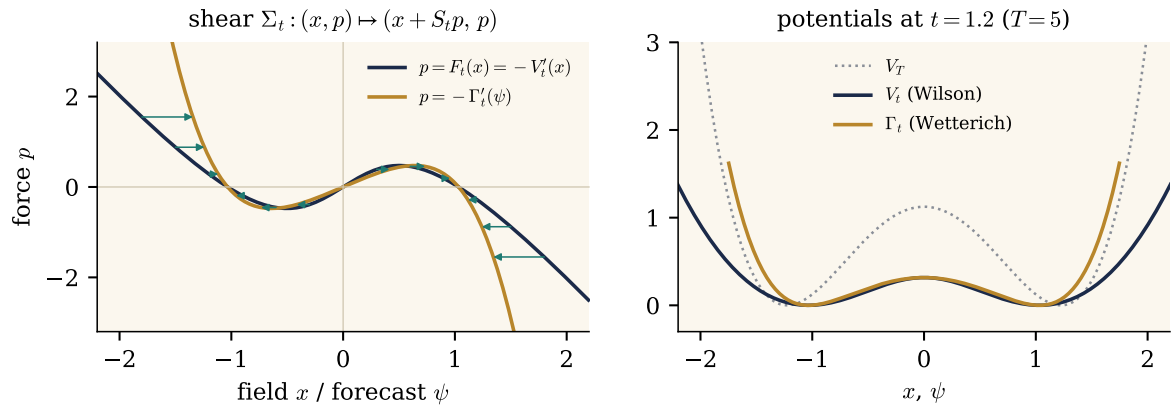


Figure 5: The zero-dimensional model of Fig. 3 at $t = 1.2$. Left: the shear Σ_t (teal arrows) maps the graph of the force $F_t = -V_t'$ (navy) to a graph over the forecast ψ (gold), which is the graph of $-\Gamma_t'$. Right: the bare potential V_T, the effective potential V_t and Γ_t, each shifted to have minimum 0.

So Γ_t, the effective average action up to the quadratic term $\frac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle$, is the free energy of the unresolved shell constrained to the forecast ψ, and (6.11) is the minimisation over the forecast. In the language of Theorem 7.1: V_t is the value function of the control problem, and Γ_t is its Moreau transform, i.e. the value function expressed in the forecast coordinate, the interaction picture of the classical part of the dynamics.

The tree level. Dropping the Itô term in (6.8) gives $\Gamma_t = V_T$, and then (6.11) becomes the Hopf–Lax formula

$$V_t^{\mathrm{tree}}(x) = \inf_y \Big\{ V_T(y) + \tfrac{1}{2}\langle y - x, S_t^{-1}(y-x)\rangle \Big\}, \tag{6.16}$$

the solution of the noiseless Polchinski equation $\partial_t V = \frac{1}{2}\langle DV, \dot{Q} DV\rangle$. It is the saddle point of the Gaussian convolution (5.3). Its minimiser solves the classical field equation $S_t^{-1}(y-x) + DV_T(y) = 0$ in the shell $[t,T]$ with background x, and expanding it in V_T produces exactly the tree diagrams.

Exercises

Exercise 6.1 (Stochastic maximum principle)**.** (a) For the control problem (7.1), perturb an optimal control $u \to u + \varepsilon v$ and show that $p_t := \mathbb{E}_t[-DV_T(X_T^u)]$ satisfies $u_t = \dot{Q}_t^{1/2} p_t$. Identify p with the force F of (5.7). (b) Show that $\mathrm{Y}_t = DF_t(\Phi_t)\dot{Q}_t^{1/2}$, and derive (6.3).

Exercise 6.2 (Characteristics)**.** (a) Show that the transport equation $\partial_t F + DF[\dot{Q}F] = 0$ is solved by $F_t(x) = F_T(y)$, $y = x + S_t F_T(y)$, and that its characteristics are the trajectories of (6.1). (b) Verify that (6.16) solves $\partial_t V = \frac{1}{2}\langle DV, \dot{Q}DV\rangle$, and that it is the limit $\hbar \to 0$ of $-\hbar \log \int \mathrm{e}^{-V_T(x+\zeta)/\hbar}\mu^{\hbar S_t}(\mathrm{d}\zeta)$. (c) For $V_T(y) = \lambda y^4$ in zero dimensions, solve the equation for y as a power series in λ, and identify the coefficients as tree graphs.

Exercise 6.3 (Conditional moments)**.** (a) Prove $\mathrm{Cov}(\mu_t^{\phi}) = S_t - S_t D^2 V_t(\phi) S_t$. (b) Prove (6.9).

Exercise 6.4 (A symplectic shear)**.** (a) Show that Σ_t preserves $\sum_i \mathrm{d}x_i \wedge \mathrm{d}p_i$ iff S_t is symmetric. (b) Show that the image of the graph of $-DV_t$ under Σ_t is the graph of $-D\Gamma_t$, with Γ_t given by (6.11). (c) For a non-gradient force, show that Theorem 6.2 still holds, but that $\mathcal{F}_t$ need not be a gradient.

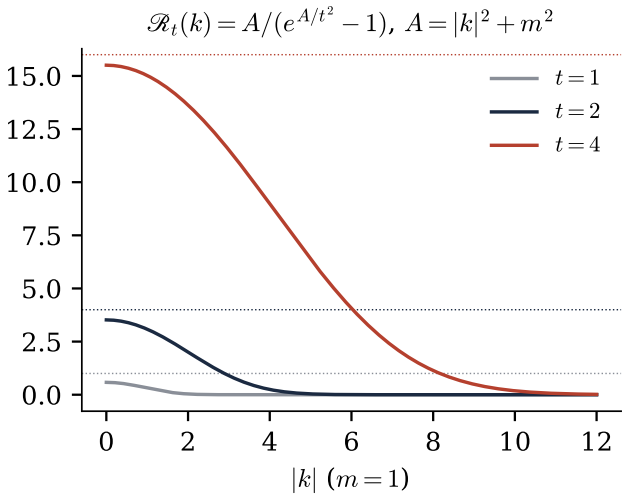


Figure 6: The regulator $\mathcal{R}_t$ of the heat profile ($T = \infty$, $m = 1$): close to the mass term t^2 (dotted) on the modes $|k| \ll t$, exponentially small on the modes $|k| \gg t$.

Exercise 6.5 (The constant)**.** Differentiate (6.11) in t at fixed x (envelope theorem), and use (5.6) to show that the quadratic terms cancel and $\partial_t \Gamma_t = -\frac{1}{2}\operatorname{Tr}(\dot{Q}_t D^2 V_t)$. Deduce (6.12) with the stated constant.

Exercise 6.6 (Gaussian check)**.** For $V_T(\varphi) = \frac{\kappa}{2}\langle \varphi, \varphi \rangle$, show that $\Gamma_t(\psi) = \frac{\kappa}{2}\langle \psi, \psi \rangle + \text{const}$, and solve the FBSDE (6.7) explicitly. Compare with Exercise 5.1.

Exercise 6.7 (1PI effective action)**.** (a) Let $\mathcal{K}(j) = \log \int \mathrm{e}^{\langle j, \varphi \rangle} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi)$. Show that $\Gamma_0^{\mathrm{EAA}} = \mathcal{K}^*$ up to a constant, and that $D^2\Gamma_0^{\mathrm{EAA}}(\psi)$ is the inverse covariance of the tilted measure with mean ψ. (b) Integrate (6.12) with $D^2\Gamma_t$ replaced by $D^2 V_T$, and obtain the one-loop formula $\Gamma_t \approx V_T + \frac{1}{2}\operatorname{Tr}\log(\mathbb{1} + S_t D^2 V_T)$.

Exercise 6.8 (Trees versus loops)**.** Using (6.11), show that $V_t = \Gamma_t - \frac{1}{2}\langle D\Gamma_t, S_t D\Gamma_t \rangle + O(\lambda^3)$ for the Φ^4 interaction (5.1). Deduce from Proposition 9.2 that, to order λ^2, Γ_t is obtained from V_t by removing exactly the term $k = 1$, the one-particle-reducible tree; the bubble ($k = 2$), the sunset ($k = 3$) and the vacuum term ($k = 4$) are 1PI and remain. Where does the sunset come from in (6.12)?

Exercise 6.9 (Regulators)**.** Derive (6.14). Compute $\mathcal{R}_t$ for the sharp Fourier profile, and for $\hat{Q}_t(k) = (|k|^2 + m^2)^{-1} - (|k|^2 + m^2 + t^2)^{-1}$ (whose kernels are not smooth, but the formulas make sense). In the last case show that $\mathcal{R}_t = t^2$ when $T = \infty$: a mass regulator.

7 Variational structure

7.1 Finite volume: the Boué–Dupuis formula

Let $\mathbb{H}_a$ be the set of progressively measurable drifts $u : [0, T] \to L^2(\mathcal{M})$ with $\mathbb{E}\int_0^T \|u_s\|^2 \mathrm{d}s < \infty$, and set

$$I_t(u) := \int_0^t \dot{Q}_s^{1/2} u_s \, \mathrm{d}s, \qquad X^u := X + I(u),$$

the field driven by the control u under the Gaussian reference $\mathbb{P}_0$.

Theorem 7.1 (Boué–Dupuis [BD98]; Polchinski as HJB)**.** (formal)

$$-\log \mathbb{E}_0\big[\mathrm{e}^{-V_T(X_T)}\big] = V_0(0) = \inf_{u\in\mathbb{H}_a} \mathbb{E}_0\Big[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s\Big]. \tag{7.1}$$

The infimum is attained at the feedback control $u_s^* = -\dot{Q}_s^{1/2} DV_s(X_s^{u^*})$*, and* $\mathrm{Law}(X_T^{u^*}) = \nu_T$*. The effective potential is the value function, and* (5.6) *is the Hamilton–Jacobi–Bellman equation of this control problem.*

Proof. (Verification.) For any u, by Itô's formula and (5.6),

$$\begin{aligned}\mathrm{d}V_t(X_t^u) &= \Big[\tfrac{1}{2}\|\dot{Q}_t^{1/2} DV_t\|^2 + \langle \dot{Q}_t^{1/2} DV_t, u_t\rangle\Big]\mathrm{d}t + \mathrm{d}M_t \\ &= \Big[\tfrac{1}{2}\|u_t + \dot{Q}_t^{1/2} DV_t\|^2 - \tfrac{1}{2}\|u_t\|^2\Big]\mathrm{d}t + \mathrm{d}M_t,\end{aligned}$$

with $DV_t = DV_t(X_t^u)$ and a martingale M. Integrating and taking expectations,

$$\mathbb{E}_0\Big[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_t\|^2 \mathrm{d}t\Big] = V_0(0) + \frac{1}{2}\mathbb{E}_0 \int_0^T \|u_t + \dot{Q}_t^{1/2} DV_t(X_t^u)\|^2 \mathrm{d}t. \tag{7.2}$$

So the cost is $\geq V_0(0)$, with equality iff $u = u^*$. With this control X^{u^*} solves the same SDE as Φ in (5.7). □

Euler–Lagrange equation = FBSDE. Perturbing $u \to u + \varepsilon v$ in (7.1), the first-order condition is $\mathbb{E}\int_0^T \langle u_s + \dot{Q}_s^{1/2}\mathbb{E}_s[DV_T(X_T^u)], v_s\rangle \mathrm{d}s = 0$ for all adapted v, i.e.

$$u_s = -\dot{Q}_s^{1/2}\,\mathbb{E}_s\big[DV_T(X_T^u)\big] = \dot{Q}_s^{1/2} F_s. \tag{7.3}$$

This is (5.8). The FBSDE is the Euler–Lagrange equation of the variational problem, and the Polchinski equation is its Hamilton–Jacobi equation. These are the three faces of the same object: *the PDE* (Polchinski), *the stochastic dynamics* (FBSDE), and *the variational principle* (Boué–Dupuis).

Entropy. By Girsanov, $\frac{1}{2}\mathbb{E}_0\int\|u\|^2 \geq \mathrm{Ent}(\mathrm{Law}(X^u) \mid \mathrm{Law}(X))$, which is at least the entropy of the terminal marginal. So (7.1) is a pathwise lift of the Gibbs variational principle $-\log Z_T = \inf_\nu\{\int V_T \,\mathrm{d}\nu + \mathrm{Ent}(\nu|\mu^{Q_T})\}$. The lift is useful because adapted drifts can be *constructed scale by scale*: one looks for u as a functional of the Gaussian process X, not as an unknown measure.

Generating functionals. Adding a test functional f,

$$\mathcal{W}_T(f) := -\log \int \mathrm{e}^{-f}\mathrm{d}\nu_T = \inf_u \mathbb{E}_0\Big[f(X_T^u) + V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s\Big] - V_0(0). \tag{7.4}$$

Bounds on $\mathcal{W}_T(f)$ uniform in T, for f in a rich class, give tightness, moment bounds and the identification of limits. For Φ_3^4 the naive problem does not have uniform bounds: the leading-order optimal drift $u_s \approx -4\lambda\dot{Q}_s^{1/2}{:}X_s^3{:}$ has a divergent cost. One renormalises by the change of variables $u = -4\lambda\dot{Q}^{1/2}{:}X^3{:} + r$. After it, and after further paracontrolled changes of variables, the singular terms are explicit polynomials in the Wick powers of X, cancelled by *the counterterms*, and the residual functional is coercive. Its control in Besov and paracontrolled norms, uniformly in T, is the construction of Φ_3^4 in [BG20]; see also [BG21].

7.2 Infinite volume

On $\mathbb{T}_L^d$ the free energy $V_0(0)$ and the effective potential V_t are extensive: they grow like L^d, and (7.1) has no limit as $L \to \infty$. Three objects are intensive and survive formally:

(i) *the force.* For a local interaction $V_T(\varphi) = \int v_T(\varphi(x))\mathrm{d}x$, $DV_T(\varphi)(x) = v_T'(\varphi(x))$, and (5.8) reads, on $\mathbb{R}^d$,

$$\Phi_t = \int_0^t \dot{Q}_s F_s \,\mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t(x) = -\mathbb{E}_t\big[v_T'(\Phi_T(x))\big]. \tag{7.5}$$

Here every quantity is local, and the equation makes sense without a volume cutoff. Infinite-volume limits of $\nu_T^{(L)}$ are expected to be laws of solutions of (7.5). Uniqueness of such solutions (e.g. for small coupling, or by contraction in weighted spaces) is a uniqueness statement for the infinite-volume theory. Non-uniqueness is expected in the presence of phase transitions, e.g. for Φ_2^4 with a deep double-well potential [GJS75];

(ii) *the Burgers flow* (5.12) of the force, for the same reason;

(iii) *differences of free energies.* For a local f, $\mathcal{W}_T^{(L)}(f)$ in (7.4) is a difference of two extensive infima, and it stays bounded as $L \to \infty$. Its variational characterisation, through the infinite-volume Euler–Lagrange equation (7.5), is the basis of the infinite-volume analysis of Φ_2^4 and $\exp(\Phi)_2$ in [BG23].

The coercive estimates must then be *local*: one uses weighted norms and needs bounds whose constants do not depend on the volume. This infrared problem is the analogue, for the scale dynamics, of the "coming down from infinity" estimates for the parabolic dynamics.

Exercises

Exercise 7.1 (Optimal drift)**.** (a) Derive (7.3) from (7.1), and check that the optimal u is a martingale up to the factor $\dot{Q}^{1/2}$. (b) For the quadratic interaction of Exercise 5.1, compute the optimal drift and the value (7.1) explicitly.

Exercise 7.2 (Cost of the leading drift)**.** For $V_T = \lambda \int :\varphi^4:_T$ on $\mathbb{T}_L^3$, show that

$$\frac{1}{2}\mathbb{E}\int_0^T \|4\lambda \dot{Q}_s^{1/2} :X_s^3:\|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4 \,\mathrm{d}z,$$

which diverges linearly in T. Relate this to the $k = 4$ term of (9.6).

Part II. Additional material

The following sections complement the main lecture. They develop the observables of the Gaussian free field and their products, the algebraic and perturbative structure of the Polchinski flow, and the Dyson–Schwinger equations at every scale.

8 Observables of the Gaussian free field and their products

8.1 Wick powers and exponentials

Point evaluations. For every x and every d, $\mathbf{X}(x) := (X_t(x))_{t\geq 0}$ is a local observable. For $d \geq 2$ it is not closed: $\mathbb{E}X_t(x)^2 = c_t \to \infty$.

Wick powers. Recall that $\partial_c H_n(a;c) = -\frac{1}{2}\partial_a^2 H_n(a;c)$ and $\partial_a H_n = nH_{n-1}$ (differentiate the generating function $\mathrm{e}^{\tau a - c\tau^2/2}$).

Proposition 8.1 (Wick observables). *For every $d \geq 1$, $n \geq 1$ and $x \in \mathcal{M}$,*

$$:X_t^n:(x) := H_n(X_t(x); c_t)$$

is a continuous martingale, with $\mathrm{d}:X_t^n:(x) = n:X_t^{n-1}:(x)\,\mathrm{d}X_t(x)$ and

$$\mathbb{E}\big[:X_t^n:(x):X_t^k:(y)\big] = \delta_{nk}\, n!\, Q_t(x,y)^n. \tag{8.1}$$

In particular, for $x \neq y$, $\lim_t \mathbb{E}[:X_t^n:(x):X_t^n:(y)] = n!\, G(x-y)^n$.

Proof. By Itô's formula, using $\mathrm{d}\langle X(x), X(x)\rangle_t = \dot{c}_t\,\mathrm{d}t$,

$$\mathrm{d}H_n(X_t(x);c_t) = \partial_a H_n\,\mathrm{d}X_t(x) + \big(\tfrac{1}{2}\partial_a^2 H_n + \partial_c H_n\big)\dot{c}_t\,\mathrm{d}t = nH_{n-1}(X_t(x);c_t)\,\mathrm{d}X_t(x).$$

The covariance is Lemma 3.8 of Lecture 2, applied to the Gaussian vector $(X_t(x), X_t(y))$. □

Compare with Theorem 3.9 of Lecture 2: there $:\varphi^n:$ existed as a random distribution only for $n(d-2) < d$, and by Proposition 3.11 not at all otherwise. As observables, Wick powers exist in *every* dimension and need no smearing. When $n(d-2) \geq d$ and $0 \leq f \not\equiv 0$, the smeared martingale $t \mapsto :X_t^n:(f)$ is unbounded in L^2 and, by the Paley–Zygmund argument of Lecture 2, not even tight as $t \to \infty$: it has no terminal value, and the observable is the whole family, not a limit. The singularity has not disappeared. It has moved from the observables themselves to their *products* at coincident points, as we now see.

Exponentials. For $\beta \in \mathbb{C}$, $:\mathrm{e}^{\beta X_t}:(x) := \exp(\beta X_t(x) - \frac{\beta^2}{2}c_t)$ is the stochastic exponential of $\beta X(x)$, hence a martingale, and $\mathbb{E}[:\mathrm{e}^{\beta_1 X_t}:(x):\mathrm{e}^{\beta_2 X_t}:(y)] = \mathrm{e}^{\beta_1\beta_2 Q_t(x,y)}$.

8.2 Products of observables

The pointwise product of two martingales is not a martingale. The natural definition of the product keeps the asymptotics.

Definition 8.2. Two observables O^1, O^2 *admit a product* if $O_t^1 O_t^2 \in L^1$ for all t and there is an observable $O^1 \star O^2$ with $\|(O^1 \star O^2)_t - O_t^1 O_t^2\|_{L^1} \to 0$. By Lemma 4.4 it is unique, and $(O^1 \star O^2)_s = \lim_t \mathbb{E}_s[O_t^1 O_t^2]$.

Proposition 8.3. *Let O^1, O^2 be continuous observables, square integrable on bounded intervals, with*

$$\mathbb{E}\int_0^\infty |\mathrm{d}\langle O^1, O^2\rangle_s| < \infty. \tag{8.2}$$

Then $O^1 \star O^2$ exists and

$$(O^1 \star O^2)_t = O_t^1 O_t^2 + \mathbb{E}_t\Big[\int_t^\infty \mathrm{d}\langle O^1, O^2\rangle_s\Big]. \tag{8.3}$$

In particular $\langle O^1 \star O^2\rangle = O_0^1 O_0^2 + \mathbb{E}\langle O^1, O^2\rangle_\infty$.

Proof. $P_t := O_t^1 O_t^2 - \langle O^1, O^2\rangle_t$ is a martingale on bounded intervals, and $N_t := \mathbb{E}_t\langle O^1, O^2\rangle_\infty$ is a closed martingale by (8.2). So $P + N$ is a martingale and equals the right-hand side of (8.3). Its distance from $O_t^1 O_t^2$ in L^1 is at most $\mathbb{E}\int_t^\infty |\mathrm{d}\langle O^1, O^2\rangle| \to 0$. □

Products of point evaluations. Here $\langle \mathbf{X}(x), \mathbf{X}(y)\rangle_t = Q_t(x-y)$ is deterministic and (8.2) reads $\int_0^\infty |\dot{Q}_s(x-y)|\mathrm{d}s < \infty$. For the heat profile and $d \geq 2$ this holds iff $x \neq y$, and then $\dot{Q}_s \geq 0$ and the integral is $G(x-y)$. Hence

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t = X_t(x)X_t(y) + G(x-y) - Q_t(x-y), \qquad \langle \mathbf{X}(x) \star \mathbf{X}(y)\rangle = G(x-y), \quad (8.4)$$

which is the correct two-point function. We never had to discuss the regularity of X_∞. For $x = y$ and $d \geq 2$ the product does not exist: its expectation would be $\lim_t \mathbb{E}X_t(x)^2 = \lim_t c_t = \infty$.

Operator product expansion. Formula (8.4) has an exact short-distance expansion at every resolution: for fixed t, as $y \to x$,

$$(\mathbf{X}(x) \star \mathbf{X}(y))_t - G(x-y) \;\longrightarrow\; X_t(x)^2 - c_t = :X_t^2:(x). \quad (8.5)$$

Since both sides are martingales, this reads $\mathbf{X}(x) \star \mathbf{X}(y) = G(x-y)\mathbb{1} + :\mathbf{X}^2:(x) + o(1)$, the remainder tending to zero at each fixed resolution t: the Wick square is the regular part of the product, and $G(x-y)$ is its singular coefficient. More generally (Exercise 8.2), for $x \neq y$,

$$:\mathbf{X}^n:(x) \star :\mathbf{X}^k:(y) = \sum_{j \leq n\wedge k} j! \binom{n}{j}\binom{k}{j} G(x-y)^j :\mathbf{X}^{n-j}(x)\mathbf{X}^{k-j}(y):, \quad (8.6)$$

where $:\mathbf{X}^p(x)\mathbf{X}^q(y):_t := H_p(X_t(x);c_t)H_q(X_t(y);c_t)$ minus its contractions at scales $\leq t$. This is Wick's theorem written at the level of observables. The coefficients $G(x-y)^j$ carry all the singularities, and the observables on the right have regular limits as $y \to x$.

Locality and factorisation. Let O^1, O^2 be observables with germs supported in $U^1_{1/t}$ and $U^2_{1/t}$, where U^i are sets at positive distance and U_r is the r-neighbourhood of U. The covariation $\mathrm{d}\langle O^1, O^2\rangle_t = \langle DO^1_t, \dot{Q}_t\, DO^2_t\rangle \mathrm{d}t$ involves $\dot{Q}_t$ only between the two supports. For a finite range decomposition it vanishes for large t, so the product exists and is asymptotically the pointwise product. For the heat profile it is exponentially small, by (4.2). Thus, under integrability conditions on the drifts of the germs, observables supported on disjoint regions can be multiplied, while the products of observables at the same point require renormalisation (Fig. 7). This is the structure of a *factorisation algebra* of observables; the precise axioms are not established here [BCG26].

Remark 8.4. A finite range decomposition makes these statements exact: if $\dot{Q}_t(x,y) = 0$ for $|x-y| \geq 1/t$, then $X_T - X_t$ restricted to regions at distance $\geq 1/t$ are independent. Such decompositions of $(m^2 - \Delta)^{-1}$ exist and are a basic tool of the rigorous renormalisation group [BBS19].

Exercises

Exercise 8.1 (Hermite calculus along the flow)**.** (a) Show that $\int H_n(a+b;c)\,\mu^s(\mathrm{d}b) = H_n(a;c-s)$ for $0 \leq s \leq c$, where $\mu^s = N(0,s)$. (b) Deduce again that $H_n(X_t(x);c_t)$ is a martingale, using only the independence of the increments.

Exercise 8.2 (Products of Wick observables)**.** (a) Show that (8.2) holds for $:\mathbf{X}^n:(x)$ and $:\mathbf{X}^k:(y)$ when $x \neq y$ and Q is the heat profile. *Hint:* c_s grows at most polynomially while $\dot{Q}_s(x-y)$ decays like $\mathrm{e}^{-s^2|x-y|^2/4}$. (b) Show that $\langle :\mathbf{X}^n:(x) \star :\mathbf{X}^n:(y)\rangle = n^2 \int_0^\infty (n-1)!\, Q_s^{n-1}\dot{Q}_s\,\mathrm{d}s = n!\, G(x-y)^n$. (c) Prove (8.6).

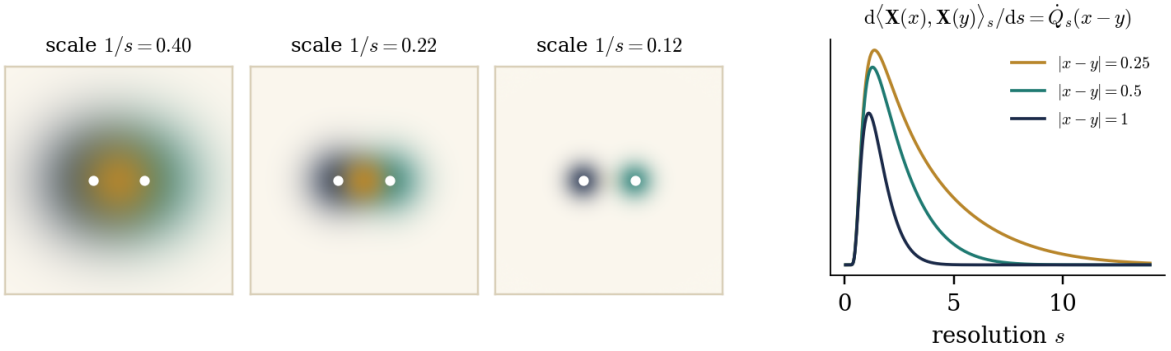


Figure 7: Covariation of point observables, heat profile in $d = 2$. Left: the kernels of $\dot{Q}_s^{1/2}$ centred at x (navy) and at y (teal) at three resolutions, with their overlap (gold). The integral of the product of the two kernels is the covariation density $\dot{Q}_s(x-y)$. Right: $s \mapsto \dot{Q}_s(x-y)$ for three distances ($m = 1$). Its integral is $G(x-y)$, and it is negligible for $s \gg 1/|x-y|$. As $y \to x$ the mass moves to $s \to \infty$, which is the divergence of the product at coincident points.

Exercise 8.3 (Exponential observables)**.** For $\beta = \mathrm{i}b$, $b \in \mathbb{R}$, and the heat profile, show that condition (8.2) for $:\mathrm{e}^{\mathrm{i}b\mathbf{X}}:(x) \star \overline{:\mathrm{e}^{\mathrm{i}b\mathbf{X}}:(y)}$ reads $\int^\infty \mathrm{e}^{b^2 c_s} \dot{Q}_s(x-y)\mathrm{d}s < \infty$. Deduce that the product exists for all $x \neq y$ if $d \leq 3$, for $|x-y| > |b|/2\pi$ but not for $|x-y| < |b|/2\pi$ if $d = 4$, and not at all if $d \geq 5$. For $d \leq 3$ show that its expectation is $\mathrm{e}^{b^2 G(x-y)}$, and decide for which b it is locally integrable in y: all b in $d = 1$, $b^2 < 4\pi$ in $d = 2$ (the L^2 threshold of Lecture 2), only $b = 0$ in $d = 3$.

9 Algebraic structure of the Polchinski flow

9.1 A semigroup of renormalisation maps

For a positive covariance S define the renormalisation map

$$\mathcal{R}_S(V) := -\log(\mu^S * \mathrm{e}^{-V}), \qquad (\mu^S * u)(\phi) = \int u(\phi + \zeta)\mu^S(\mathrm{d}\zeta). \tag{9.1}$$

Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T). \tag{9.2}$$

So the effective potentials are an orbit of an *abelian* semigroup, indexed by the cone of positive covariances, acting nonlinearly on interactions and linearly on Boltzmann factors e^{-V}. Its infinitesimal generators are the vector fields

$$\mathcal{P}_B(V) := \tfrac{1}{2}\operatorname{Tr}(BD^2V) - \tfrac{1}{2}\langle DV, BDV\rangle, \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\mathcal{R}_{sB}(V)\big|_{s=0} = \mathcal{P}_B(V), \tag{9.3}$$

and (5.6) reads $\partial_t V_t = -\mathcal{P}_{\dot{Q}_t}(V_t)$.

Proposition 9.1 (Flatness)**.** (formal) *(a) $[\mathcal{P}_B, \mathcal{P}_{B'}] = 0$ for all positive B, B'. (b) V_t depends on the profile only through $Q_{T,t}$. In particular the free energy $V_0(0)$ and the law ν_T depend only on Q_T, not on the path $t \mapsto Q_t$.*

Proof. (a) Differentiate $\mathcal{R}_{sB}\mathcal{R}_{rB'} = \mathcal{R}_{rB'}\mathcal{R}_{sB}$ at $s = r = 0$, or compute directly (Exercise 9.1). (b) is (9.2). □

Different profiles with the same endpoints are different *coordinate systems* on the same renormalisation group trajectory. They change the process Φ and the observables at intermediate resolution, but not the terminal theory. The flow has no holonomy [BCG26].

9.2 Perturbation theory and counterterms

Expanding $V_t = \mathcal{R}_{Q_{T,t}}(V_T)$ in cumulants of the Gaussian increment $\zeta \sim \mu^{Q_{T,t}}$ gives

$$V_t(\phi) = \mathbb{E}_\zeta\big[V_T(\phi+\zeta)\big] - \tfrac{1}{2}\operatorname{Var}_\zeta\big(V_T(\phi+\zeta)\big) + \dots \tag{9.4}$$

For (5.1), Exercise 8.1 shows that Gaussian convolution simply *lowers the Wick ordering*: since $Q_{T,t}(x,x) = c_T - c_t$,

$$V_t(\phi) = \int \big[\lambda H_4(\phi(x); c_t) + \gamma_T H_2(\phi(x); c_t)\big]\mathrm{d}x + O(\lambda^2). \tag{9.5}$$

Here $\gamma_T = O(\lambda^2)$ is kept for later use. At first order the effective potential is the Wick-ordered potential at the current resolution. This is the dual statement of Proposition 8.1: $\int H_4(X_t; c_t)$ is an observable.

Proposition 9.2 (Second order)**.** (formal) *Write $S = Q_{T,t}$ and $:\phi^p:_t = H_p(\phi; c_t)$. Then*

$$V_t(\phi) = \int \big[\lambda{:}\phi^4{:}_t + \gamma_T{:}\phi^2{:}_t\big] - \frac{\lambda^2}{2}\sum_{k=1}^{4}\binom{4}{k}^2 k! \iint S(x,y)^k\, {:}\phi^{4-k}{:}_t(x)\, {:}\phi^{4-k}{:}_t(y)\,\mathrm{d}x\,\mathrm{d}y + O(\lambda^3). \tag{9.6}$$

Proof. By the binomial formula for Hermite polynomials,

$$H_4(\phi+\zeta; c_t + s) = \sum_{k=0}^{4}\binom{4}{k} H_k(\zeta; s) H_{4-k}(\phi; c_t), \qquad s = S(x,x).$$

Then compute the variance in (9.4) with Lemma 3.8 of Lecture 2. □

Now let $T \to \infty$ at fixed t, so that $S \to C - Q_t$, which near the diagonal is as singular as G. In $d = 2$ every S^k is locally integrable, so with $\gamma_T = 0$ every term has a limit: Wick ordering suffices. In $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1, 2$: S and S^2 are locally integrable. These terms are finite and nonlocal.
- $k = 3$ (the *sunset*): $S^3 \sim |z|^{-3}$ is not integrable. Writing $\phi(x)\phi(y) = \frac{1}{2}(\phi(x)^2 + \phi(y)^2) - \frac{1}{2}(\phi(x) - \phi(y))^2$, up to a field-independent constant the divergent part is local: $-48\lambda^2\big(\int S^3\big)\int {:}\phi^2{:}_t + (\text{finite})$, with $\int S^3 \sim \frac{1}{16\pi^2}\log T$. It is cancelled by

$$\gamma_T := 48\lambda^2 \int_{\mathbb{T}_L^3} Q_T(z)^3\,\mathrm{d}z = \frac{3\lambda^2}{\pi^2}\log T + O(1), \tag{9.7}$$

  since $\gamma_T - 48\lambda^2\int S^3$ converges (Exercise 9.2).
- $k = 4$: a divergent constant $-12\lambda^2 L^3 \int S^4 \sim T$, the vacuum energy. It cancels in ν_T but not in the free energy.

Power counting (Exercise 9.3) shows that no further field-dependent counterterms are needed in $d = 3$: Φ^4_3 is superrenormalisable. Thus renormalisation appears as a property of the flow: one chooses the terminal data V_T so that the effective potentials V_t at *fixed* resolution have limits. The divergences at second order are also related to the singularity of Φ^4_3 with respect to the GFF [BG21]: the measures ν_t at fixed t are expected to stay absolutely continuous with respect to μ^{Q_t}, while the terminal one is not.

Exercises

Exercise 9.1 (Commuting flows)**.** Show by direct computation that $D\mathcal{P}_B(V)[\mathcal{P}_{B'}(V)] = D\mathcal{P}_{B'}(V)[\mathcal{P}_B(V)]$. Which property of D^2V and D^3V is used?

Exercise 9.2 (The sunset)**.** (a) Show that $\int_{|z|\le 1} G(z)^3\,\mathrm{d}z = \infty$ in $d = 3$, and that $\int Q_T^3 = \frac{1}{16\pi^2}\log T + O(1)$ for the heat profile. (b) Show that $Q_T^3 - (Q_T - Q_t)^3$ is integrable uniformly in T, and deduce that the counterterm (9.7) makes the $:\phi^2:$ coefficient in (9.6) converge. (c) Show that the remaining part $\iint S^3(x,y)(\phi(x)-\phi(y))^2$ converges for smooth ϕ.

Exercise 9.3 (Power counting)**.** Show that the superficial degree of divergence of a connected graph of Φ^4_d with n vertices and E external legs is $\omega = d - \frac{d-2}{2}E + (d-4)n$. Deduce that in $d = 3$ the only divergent graphs with loops are the two-point graphs with $n \le 2$ (the tadpole and double tadpole, removed by Wick ordering, and the sunset) and vacuum graphs with $n \le 3$.

10 Dyson–Schwinger equations at every scale

Assume now that all the operators involved are functions of A, hence commute, and let

$$K := Q_T^{-1}, \qquad J_t := KQ_t, \qquad J_0 = 0, \quad J_T = \mathbb{1}.$$

For the heat profile, $K = Ae^{A/T^2}$ and $J_t = \mathrm{e}^{-A(t^{-2}-T^{-2})}$ is a heat semigroup, which tends to e^{-A/t^2} as $T \to \infty$. F_t is the force observable of Theorem 5.4.

Theorem 10.1 (DS equations at resolution t)**.** (formal) *The* noise observable

$$\eta_t := K\Phi_t - J_tF_t \tag{10.1}$$

is a centred $\mathbb{P}$-martingale, with $\mathrm{d}\eta_t = K\dot{Q}_t^{1/2}\mathrm{d}W_t^{\mathbb{P}} - J_t\,\mathrm{d}F_t$. For every $t \le T$, every h and every functional G,

$$\mathbb{E}^{\mathbb{P}}\left[D_{J_th}G(\Phi_t)\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, \eta_t\rangle\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, K\Phi_t - J_tF_t\rangle\right]. \tag{10.2}$$

At $t = T$ this is the Dyson–Schwinger equation of ν_T,

$$\mathbb{E}^{\nu_T}[D_hG] = \mathbb{E}^{\nu_T}\left[G\,\langle h, K\varphi + DV_T(\varphi)\rangle\right]. \tag{10.3}$$

Proof. From (5.7) and $\dot{J}_t = K\dot{Q}_t$, $\mathrm{d}\eta_t = K(\dot{Q}_tF_t\mathrm{d}t + \dot{Q}_t^{1/2}\mathrm{d}W_t^{\mathbb{P}}) - K\dot{Q}_tF_t\mathrm{d}t - J_t\mathrm{d}F_t$. The drift cancels, F is a martingale and $\eta_0 = 0$. For (10.2) use Proposition 5.2: $\Phi_t \sim \nu_t \propto \mathrm{e}^{-V_t}\mu^{Q_t}$. Gaussian integration by parts for μ^{Q_t} in the direction Q_tk gives $\mathbb{E}^{\nu_t}[D_{Q_tk}G] = \mathbb{E}^{\nu_t}[G(\langle k, \varphi\rangle + \langle Q_tk, DV_t(\varphi)\rangle)]$. Take $k = Kh$, so that $Q_tk = J_th$, and recall $F_t = -DV_t(\Phi_t)$. □

Formally letting $T \to \infty$, $K \to A$, and (10.1) becomes an exact identity between observables at every resolution:

$$A\Phi_t = AQ_t\,F_t + \eta_t, \tag{10.4}$$

with $AQ_t = \mathrm{e}^{-A/t^2}$ for the heat profile. This is the Euclidean field equation $A\varphi = -V'(\varphi) +$ noise of Lectures 1–2, seen at resolution t. It needs no products of distributions: the nonlinearity enters only through the force *observable* F_t, smeared at scale $1/t$. For the GFF ($F = 0$), (10.2) is Gaussian integration by parts for μ^{Q_t}. For Φ^4_3, where F_t plays the role of the renormalised cube, the limit $t \to \infty$ of (10.4) corresponds to the renormalised DS equation of [GH21].

Remark 10.2 (Comparison with stochastic quantisation). In parabolic or elliptic stochastic quantisation (Lecture 2, §3.5–3.6) the field equation is imposed in an auxiliary time, and the field at fixed time has the right law. Here the field equation (10.4) holds at each resolution, and the auxiliary parameter is the resolution itself. Taking conditional expectations of an SPDE solution with respect to a scale filtration gives identities of the same shape, but with a different noise covariance: the martingale $J_t\,\mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential [BCG26].

Exercises

Exercise 10.1 (DS from the dynamics)**.** Prove (10.2) without using Proposition 5.2, for G of the form $G_t(\Phi_t)$ with G_t solving (5.10). Show that the drift of $D_{J_t h}G_t(\Phi_t)$ equals the covariation of $G_t(\Phi_t)$ with $\langle h, \eta_t\rangle$. Where is the symmetry of DF_t used?

Exercise 10.2 (Gaussian check)**.** For $V_T = 0$ show that (10.2) reduces to $\mathbb{E}[D_{Q_t Kh}G(X_t)] = \mathbb{E}[G(X_t)\langle Kh, X_t\rangle]$, and compare with the DS characterisation of the GFF in Lecture 2.

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